

GaaS Workload Characterization under NUMA Architecture for Virtualized GPU

Huixiang Chen, Meng Wang, Yang Hu, Mingcong Song, Tao Li

Presented by **Huixiang Chen**

ISPASS 2017

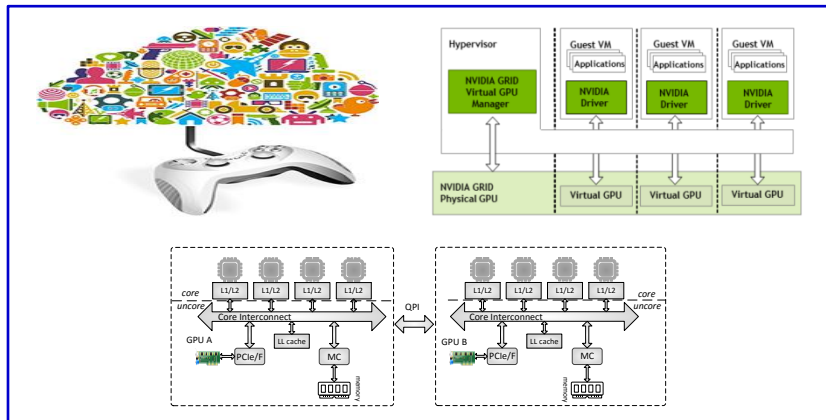
April 24, 2017, Santa Rosa, California

**IDEAL (Intelligent Design of Efficient Architectures Laboratory)
Department of Electrical and Computer Engineering
University of Florida**

Talk Overview



1. Background and Motivation

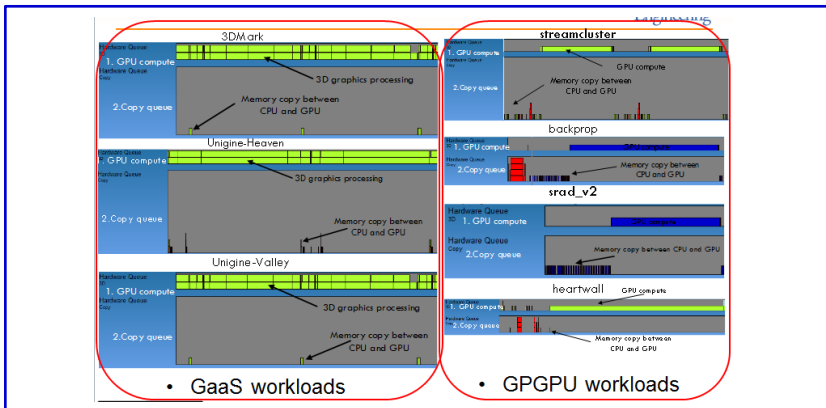


2. Experiment Setup

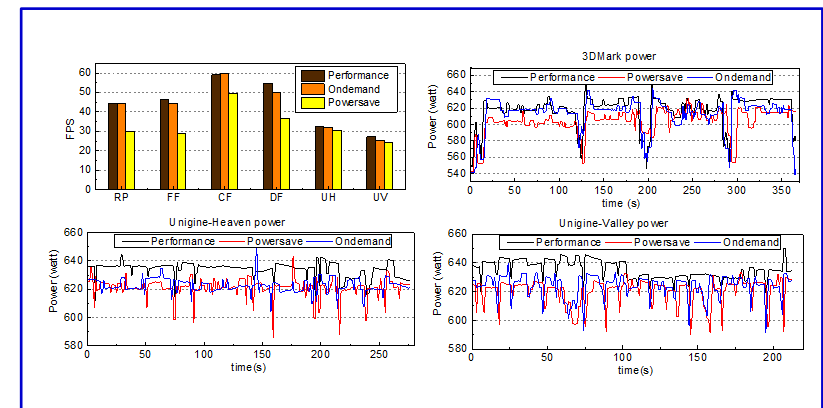


	Physical GPUs	VGPU type	Frame Buffer (Mbytes)	Maximum vGPUs per GPU
GR ID K2	2	K280	4096	1
		K260	2048	2
		K240	1024	4
		K220	512	8

3. Characterizations and Analysis



4. DVFS



Graphics-as-a-Service (GaaS)



Cloud Gaming



Video Streaming



Virtual Desktop (VDI)



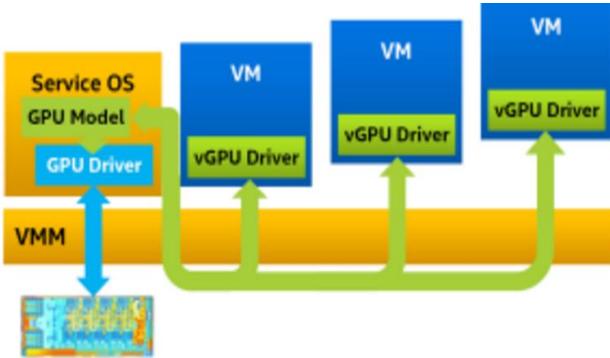
GPU Virtualization!

- 1. API Intercept
- 2. GPU pass-through
- 3. Shared virtualized GPU

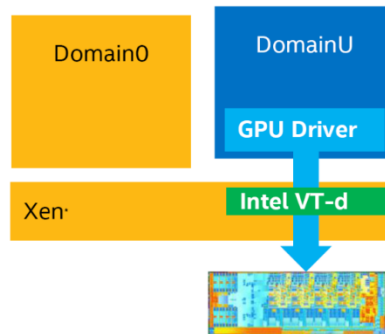
GPU Virtualization



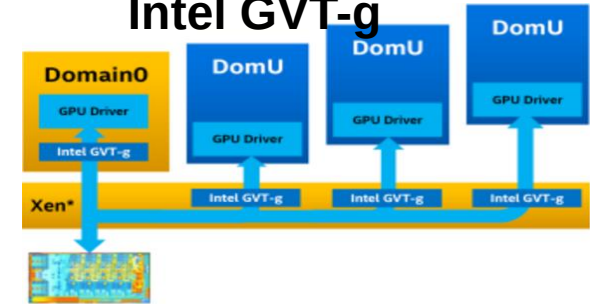
Intel GVT-s



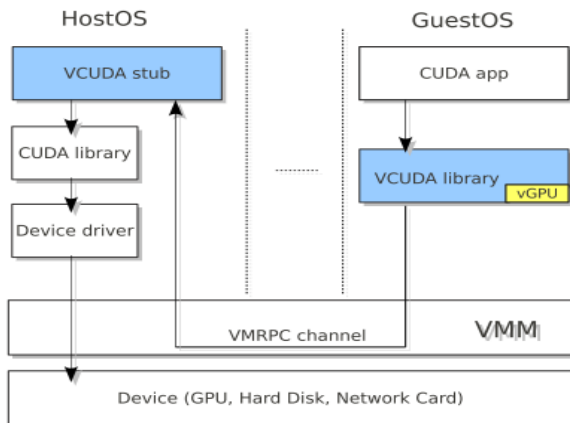
Intel GVT-d



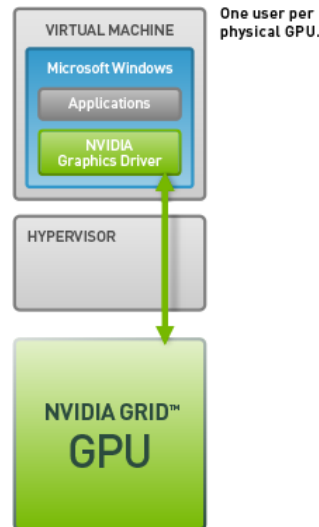
Intel GVT-g



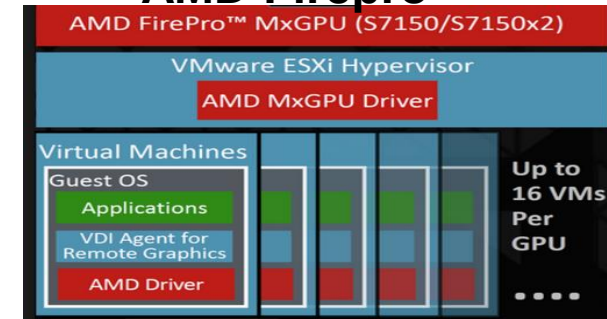
vCUDA



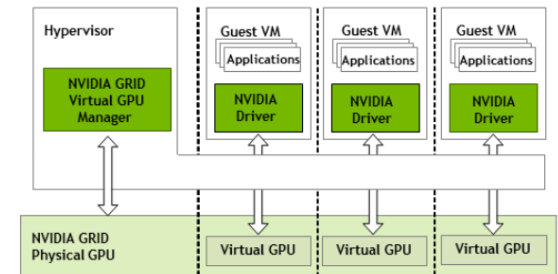
NVIDIA GPU-passthrough



AMD Firepro



NVIDIA GRID

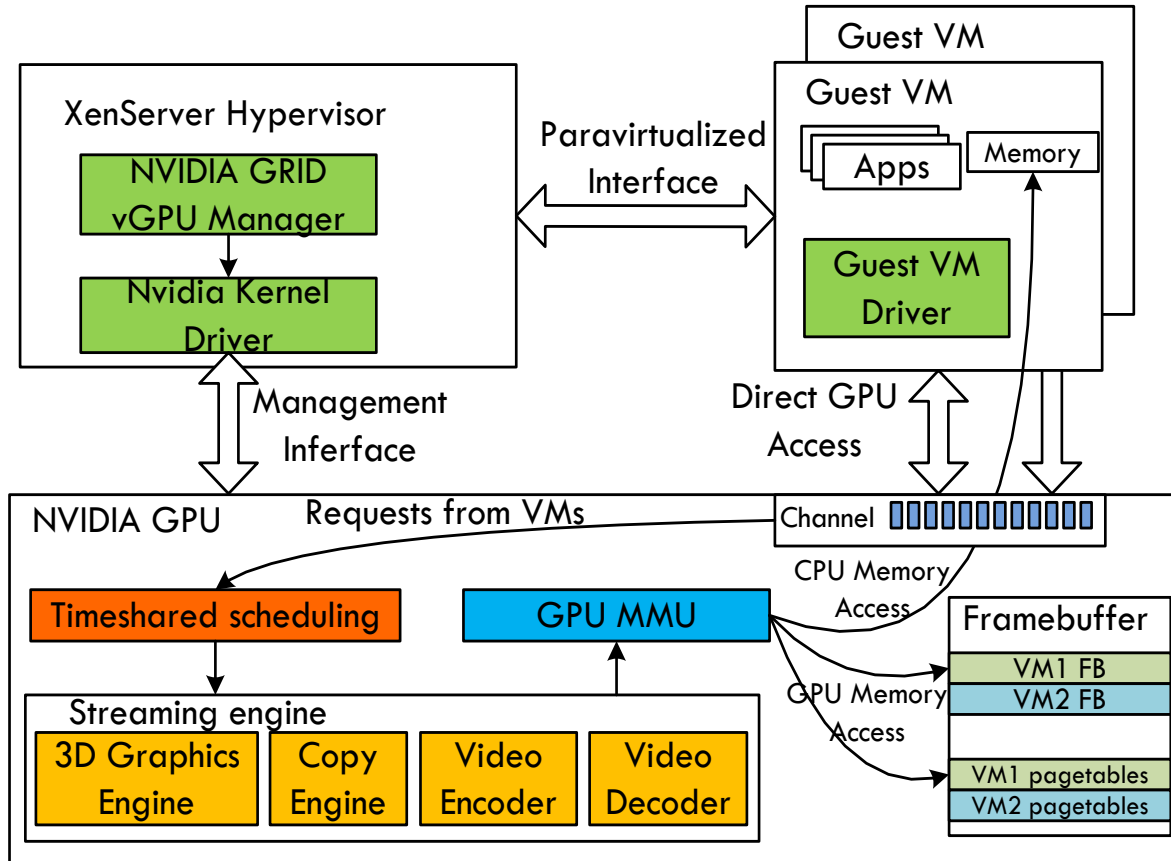


1. API intercept

2. GPU pass-through

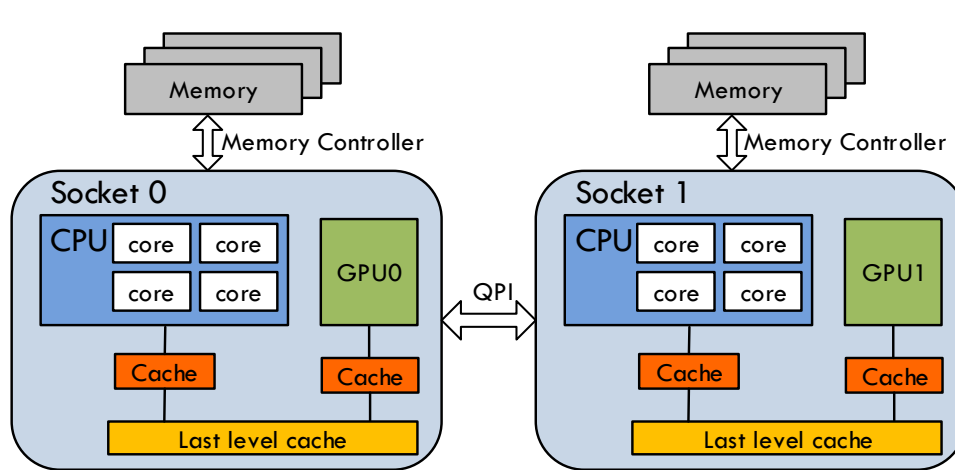
3. Virtualized GPU

NVIDIA GRID GPU Virtualization

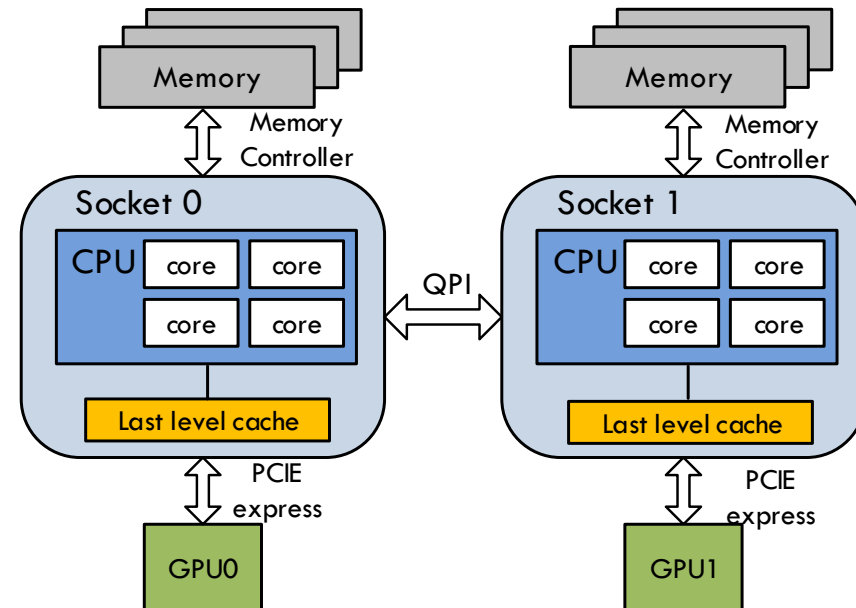


GPU NUMA issue

- Unified Architecture
- Discrete Architecture

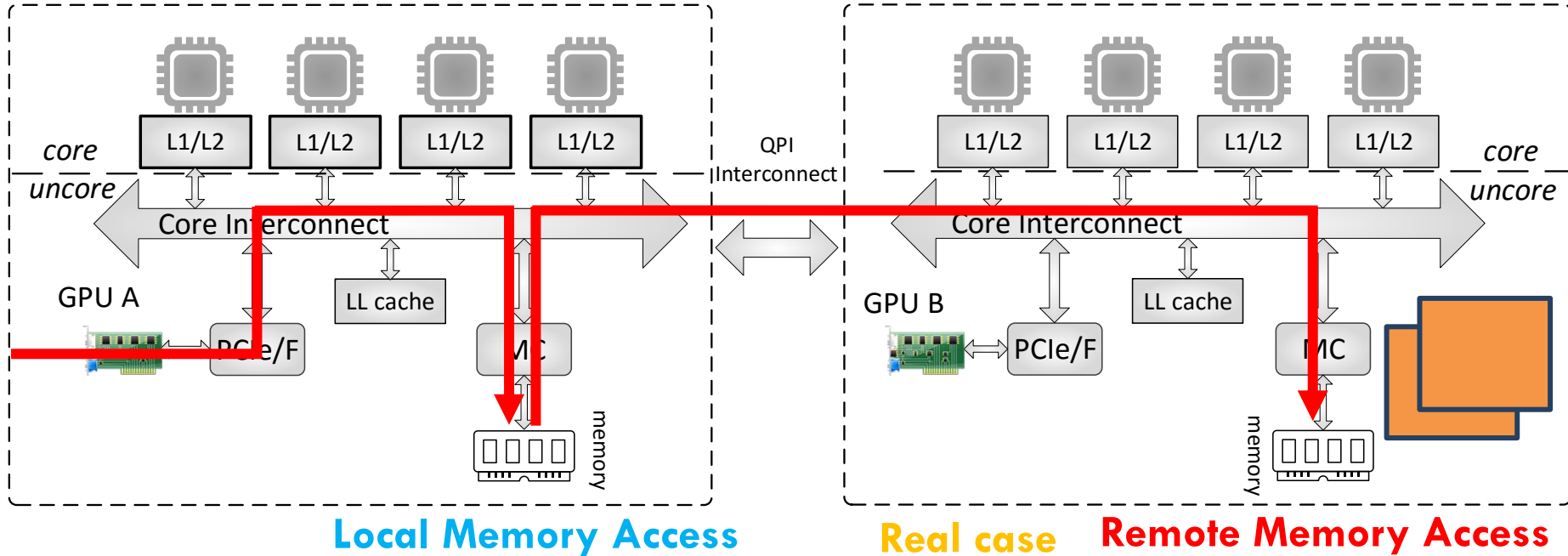


Unified Architecture

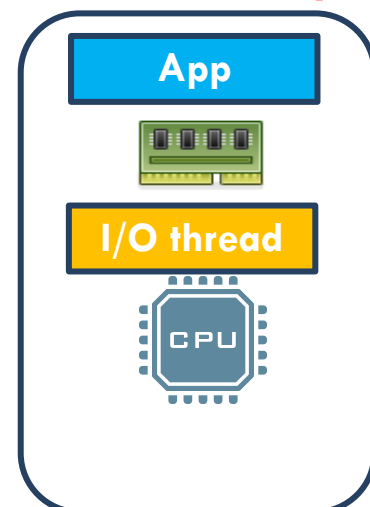
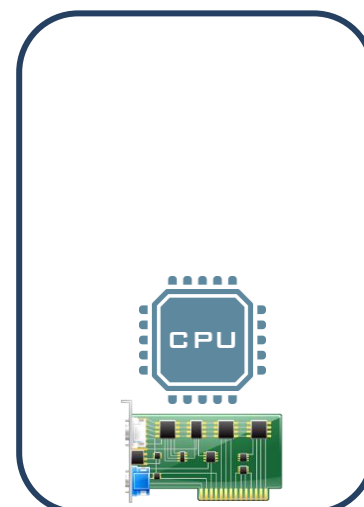
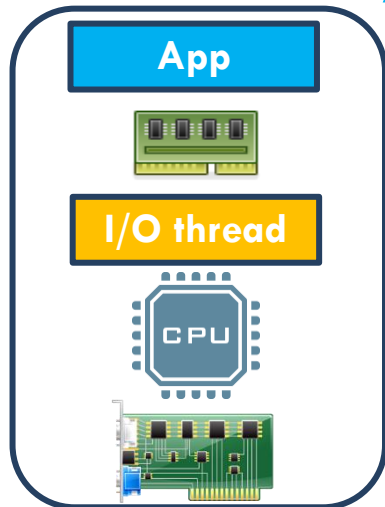


Discrete Architecture

GPU NUMA Issue



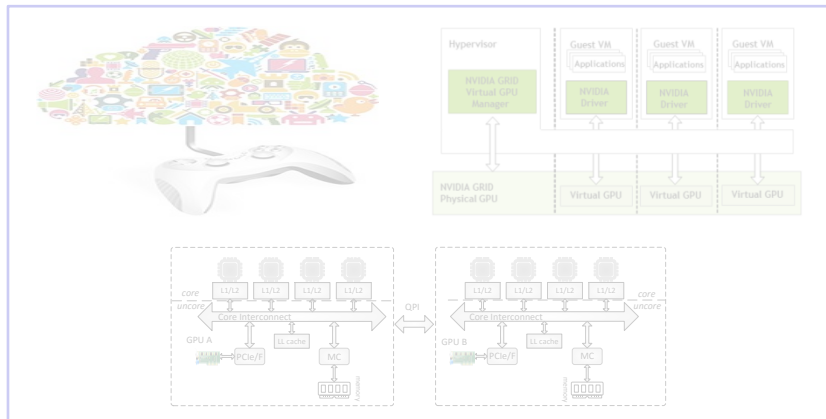
Ideal



Talk Overview



1. Background and motivation

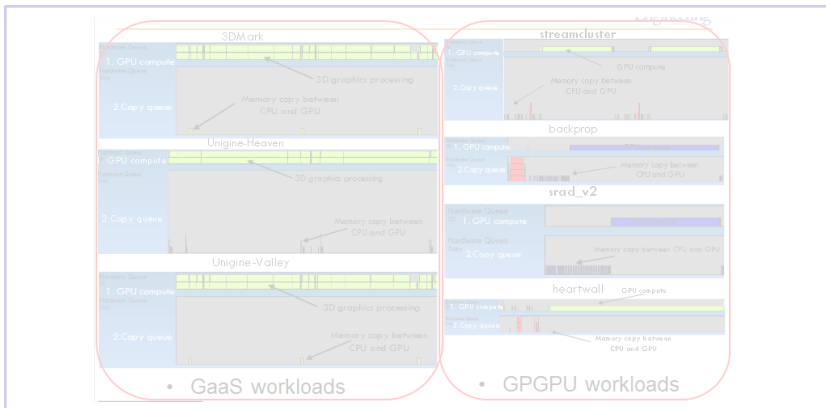


2. Experiment Setup

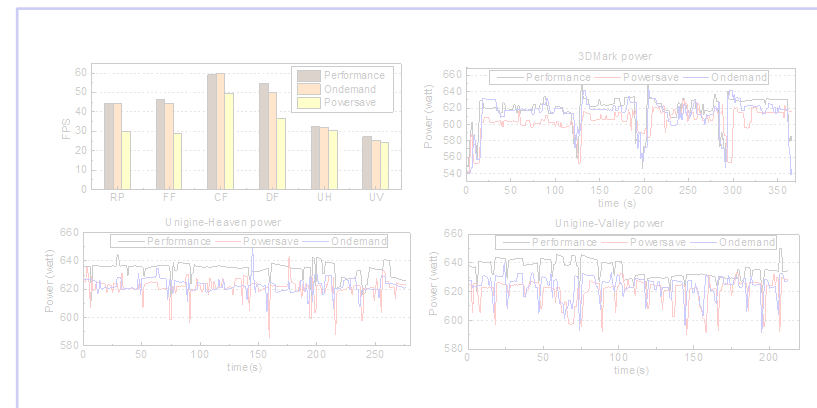


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3. Characterizations and Analysis



4. DVFS



Experiment Setup



- **Platform Configuration**
 - 4U Supermicro Server
 - XenServer 7.0
 - Intel QPI, 6.4 GT/s
 - NVIDIA GRID K2, 8GB GDDR5, 225W, PCIE 3.0 x 16

	Physical GPUs	VGPU type	Frame Buffer (Mbytes)	Maximum vGPUs per GPU
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- **Workloads and Metrics**

- **GaaS workloads:** Unigine-Heaven, Unigine-Valley, 3DMark (Return to Proxycon, Firefly Forest, Canyon Fly, Deep Freeze)
- Performance metrics: frame-per-seconds (FPS)
- **GPGPU workloads:** Rodinia benchmark
- Performance metrics: execution time

- **Local mapping:**

- the Guest VM's vCPUs are statically pinned to the local socket close to the GPU.

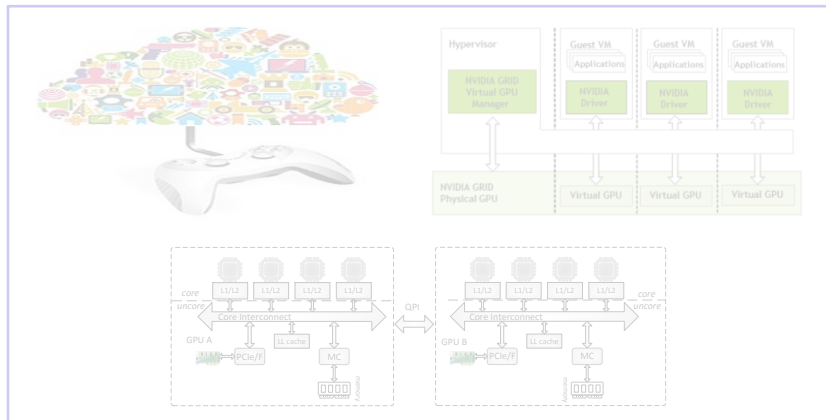
- **Remote mapping:**

- the vCPUs are statically pinned to the remote socket. (XenServer controls the memory affinity automatically close to the CPU affinity).

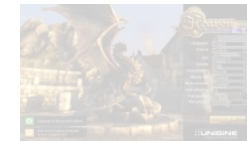
Talk Overview



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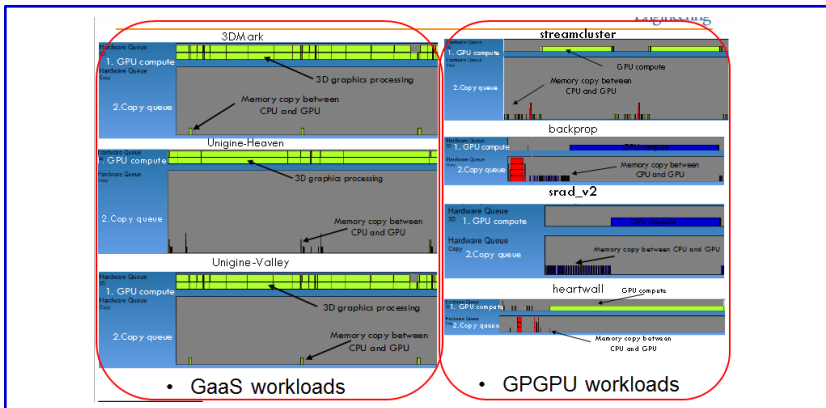


2. Experiment Setup

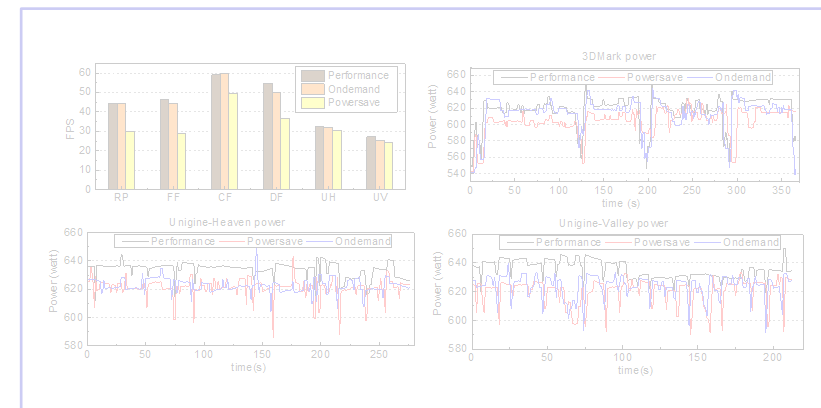


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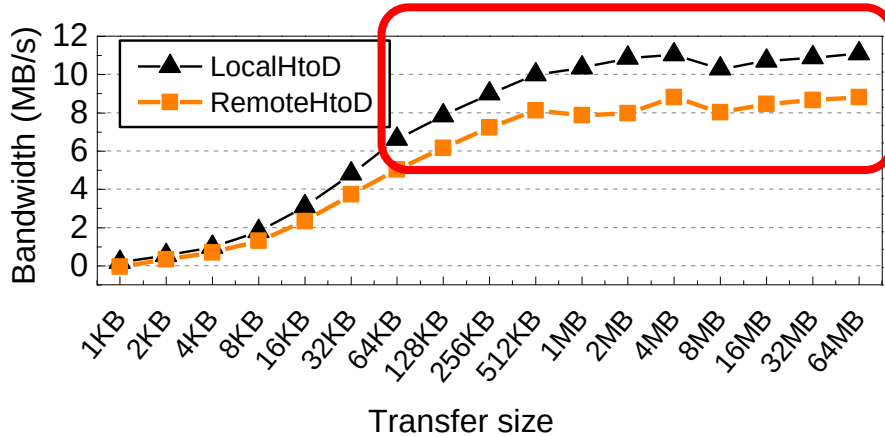
3. Characterizations and Analysis



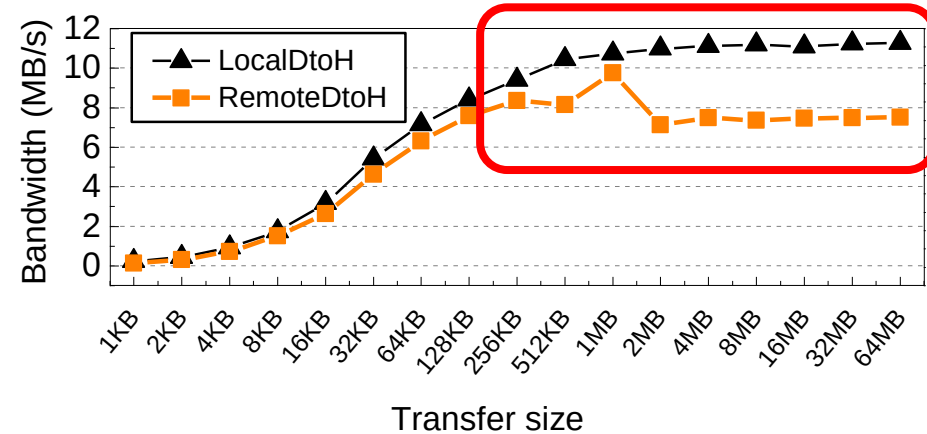
4. DVFS



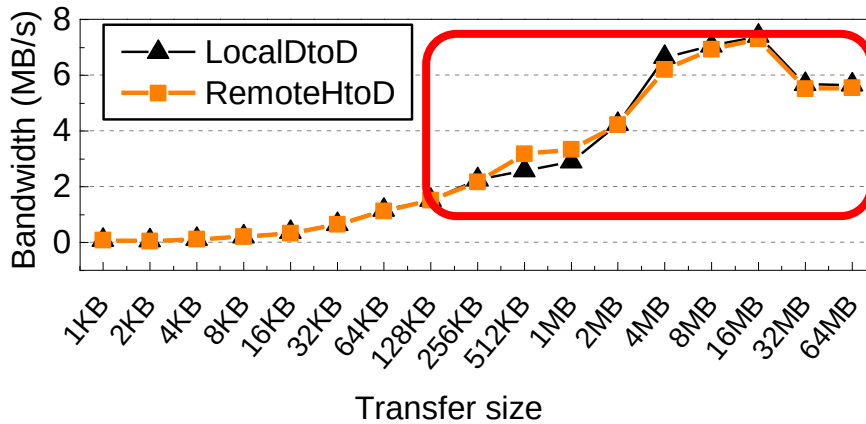
NUMA Transfer Bandwidth



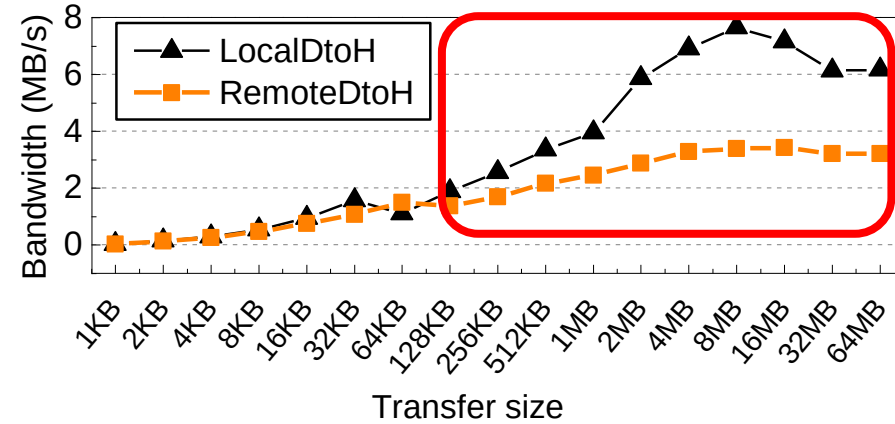
CPU→GPU, pinned memory



GPU→CPU, pinned memory



CPU→GPU, pageable memory



GPU→CPU, pageable memory

NUMA Transfer Bandwidth

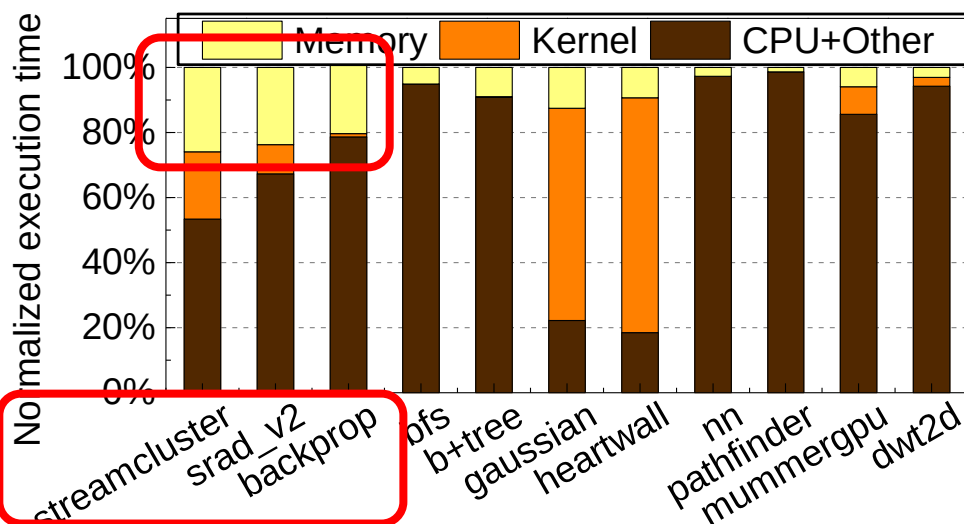
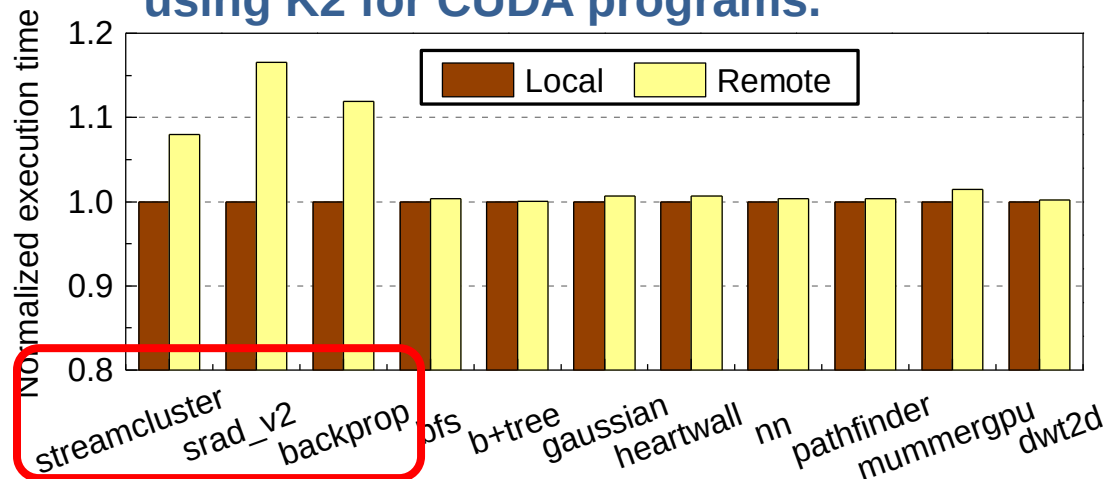


- Pinned memory:
 - 10% NUMA overhead for writing data to GPU, 20% reading data back from GPU
- Pageable memory:
 - close to 0 NUMA overhead for writing, 50% for reading data back from GPU

NUMA Performance Difference-GPGPU Workloads



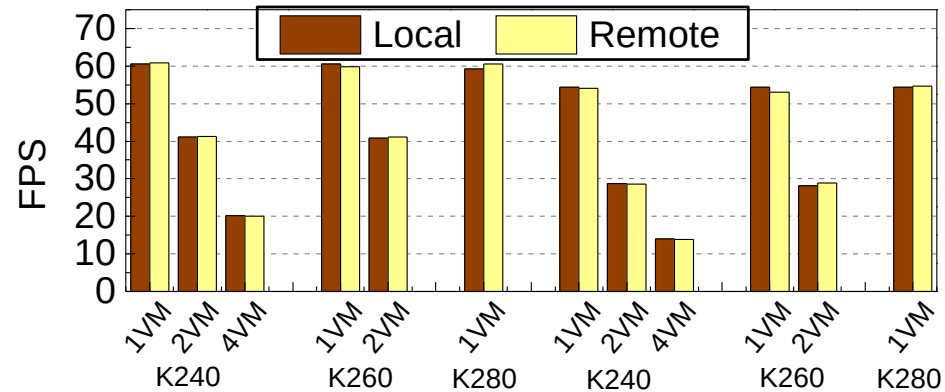
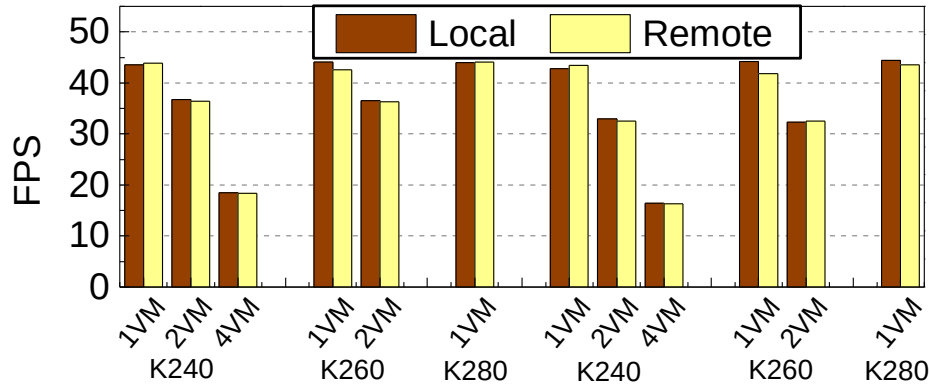
- Note: only 1VM can be configured using K2 for CUDA programs.



Remarks

- For GPGPU workloads *streamcluster*, *srاد_v2*, *backprop* stands out
- Further breakdown shows that for GPGPU workloads, the more time spent on CPU GPU communication, the higher NUMA overhead there is.

NUMA Performance Difference-GaaS Workloads



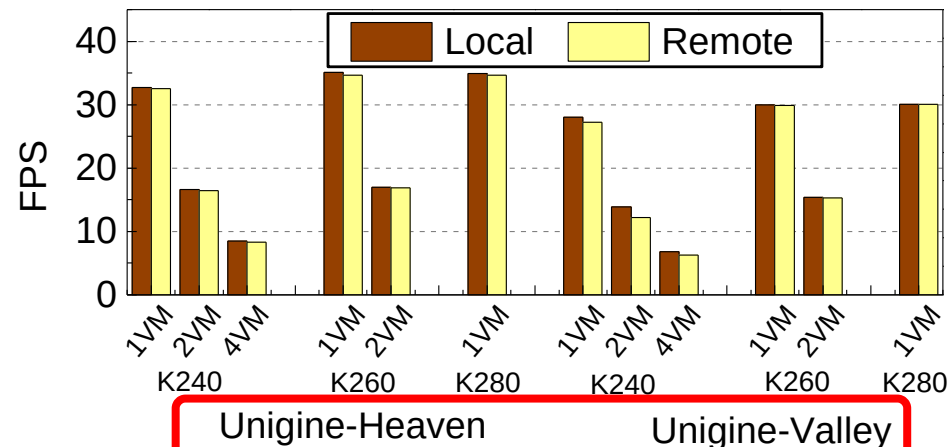
3DMark

Return to Proxycon

Firefly Forest

Canyon Flight

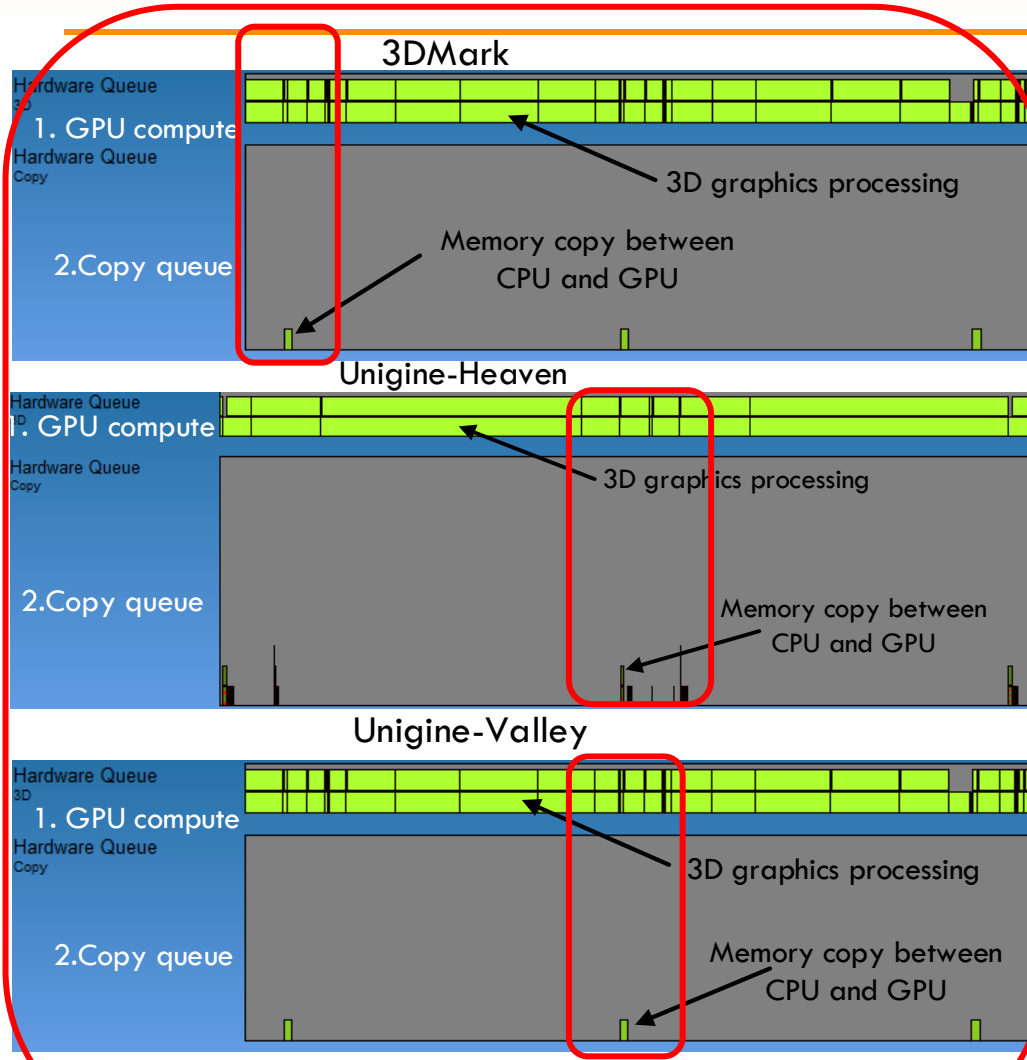
Deep Freeze



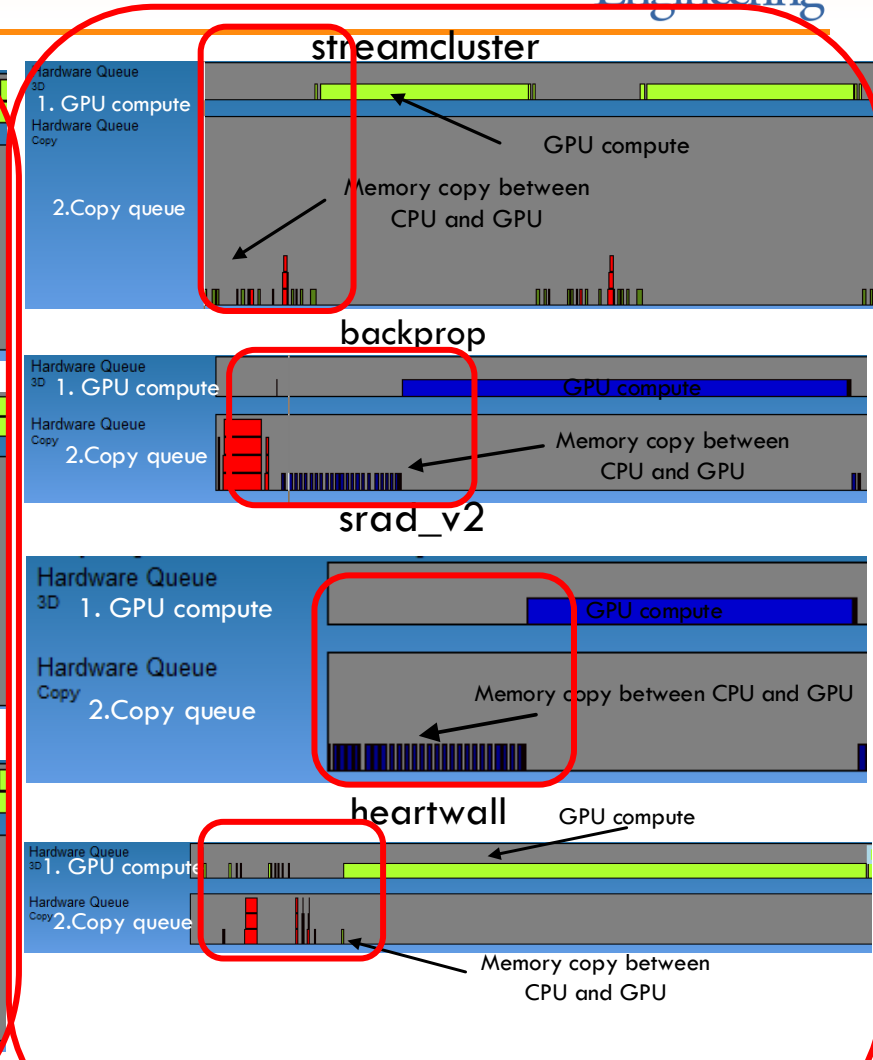
- GaaS workloads

- Little NUMA overhead exists

GaaS Overhead Analysis Cont. (1)



- GaaS workloads



- GPGPU workloads

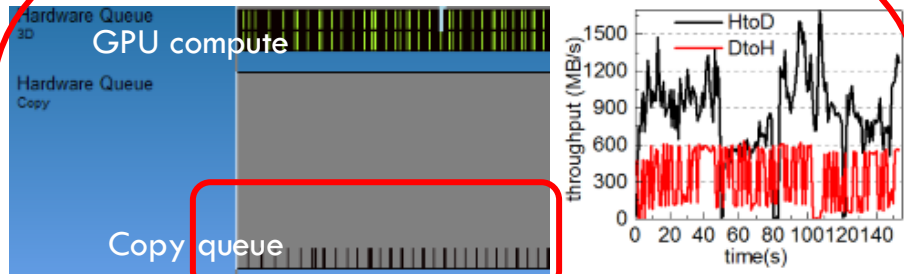
GaaS Overhead Analysis Cont. (1)



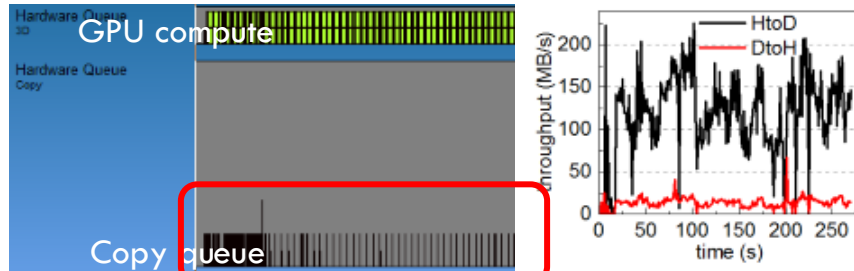
- 1. For GaaS workloads, most memory copy operations between CPU and GPU are overlapped with graphics processing operations. However, GPGPU workloads are different. Little overlap happens.
- 2. The communication time is trivial compared to GPU computing in the graphics queue, which clearly shows the GPU-computation intensive feature.

GaaS Overhead Analysis Cont. (2)

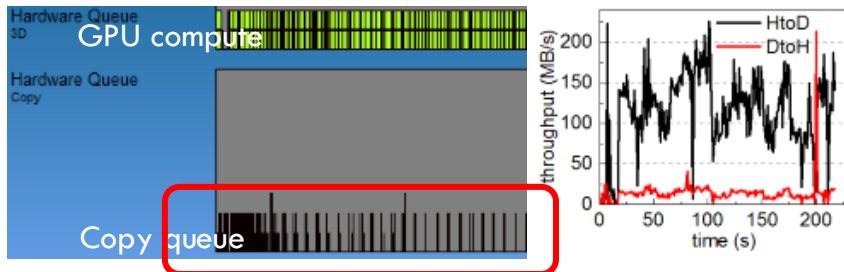
3DMark



Unigine-Heaven

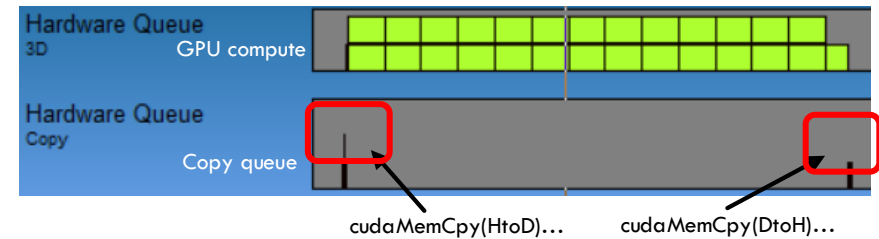


Unigine-Valley



- GaaS workloads

hearwall

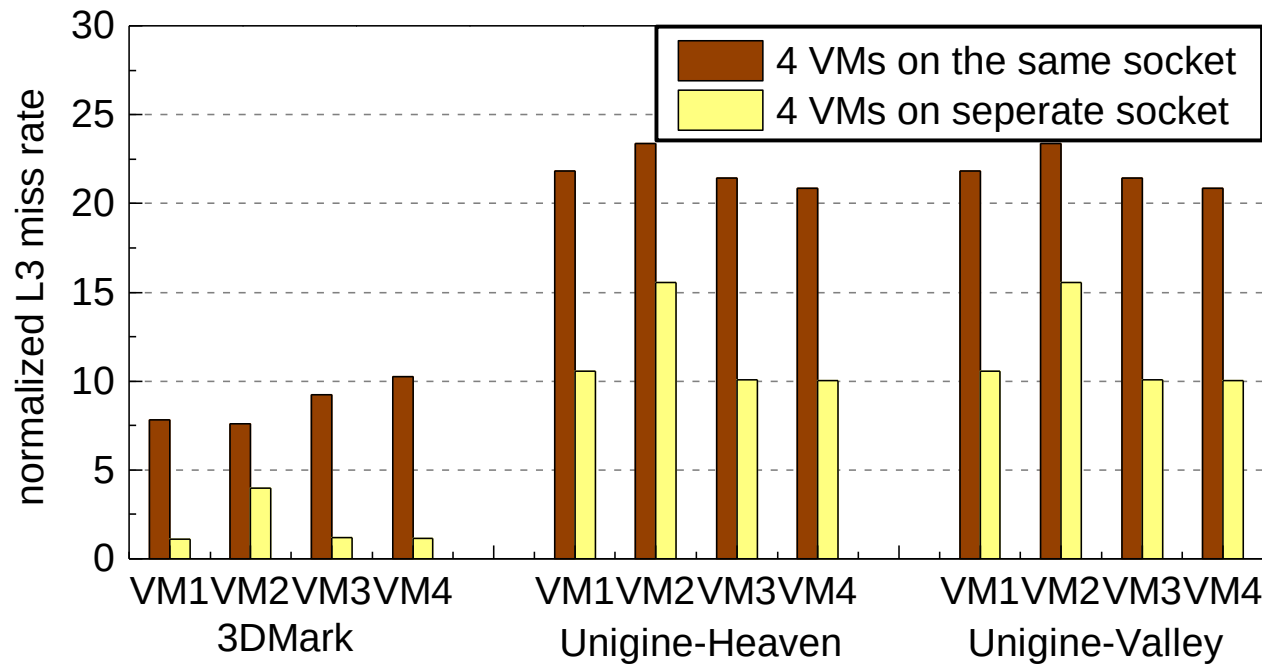


GaaS Overhead Analysis Cont. (2)



- GaaS workloads incurs more real-time processing, compared with GPGPU workloads. This kind of workload behavior makes it easier for memory transfers overlapping with GPU computing.

Influence of CPU uncore

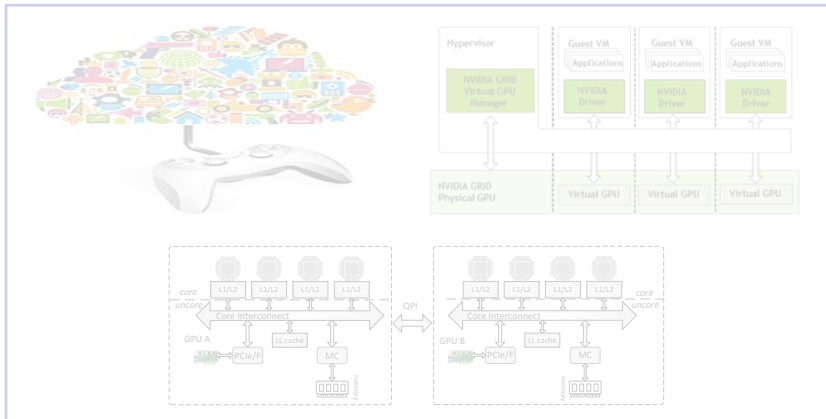


- CPU uncore has little performance influence on GPU NUMA for GaaS

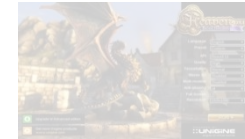
Talk Overview



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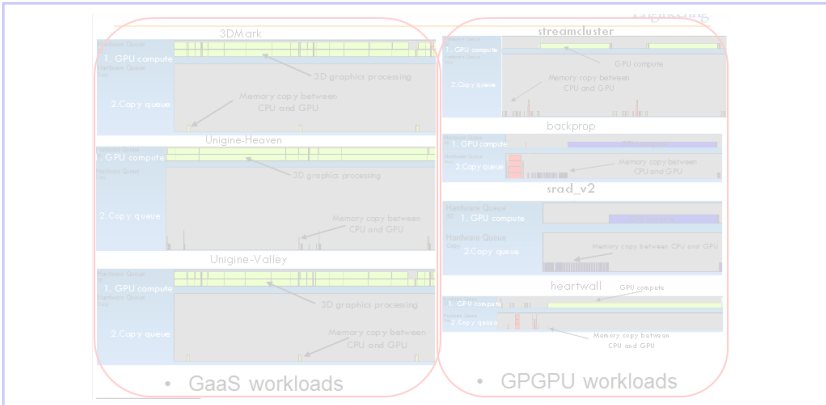


2. Experiment Setup

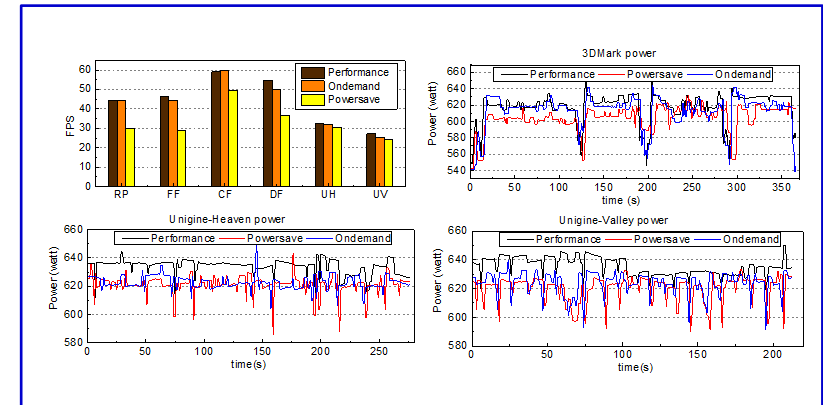


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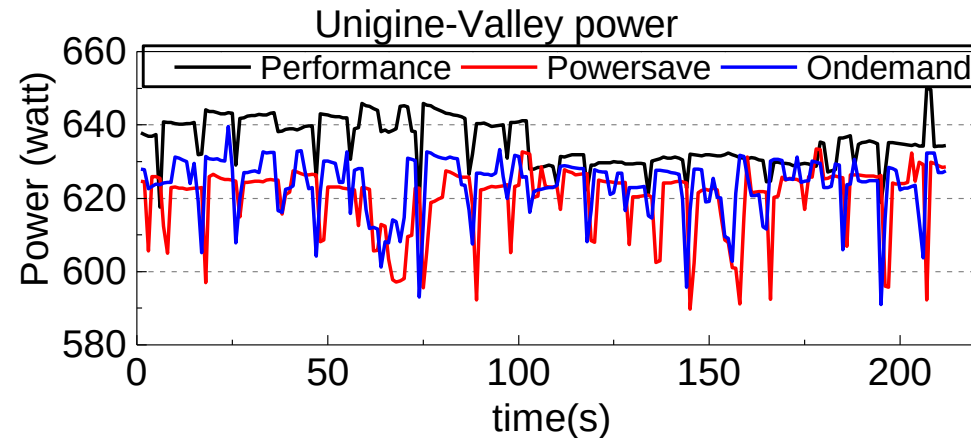
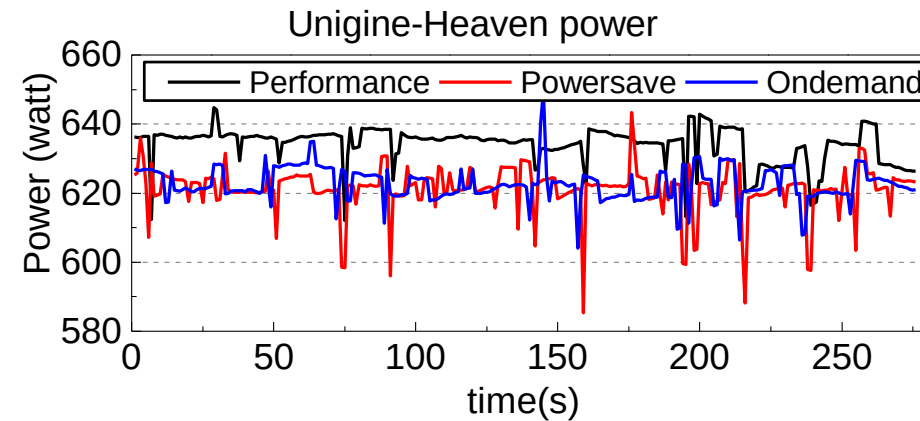
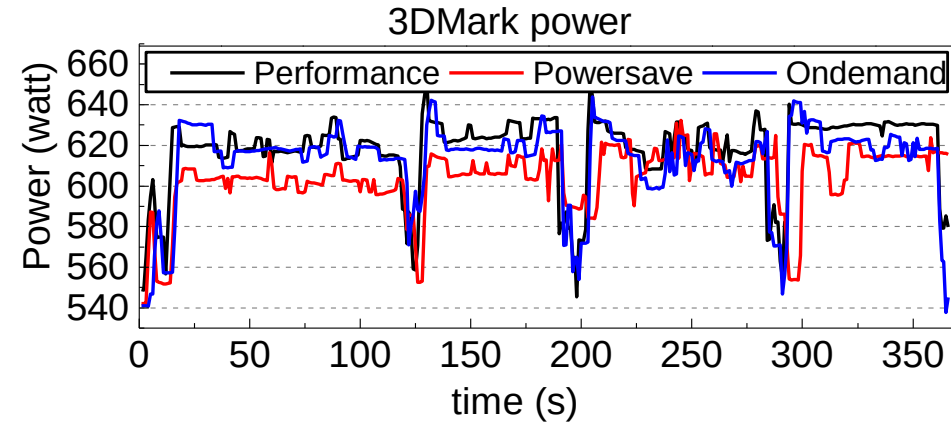
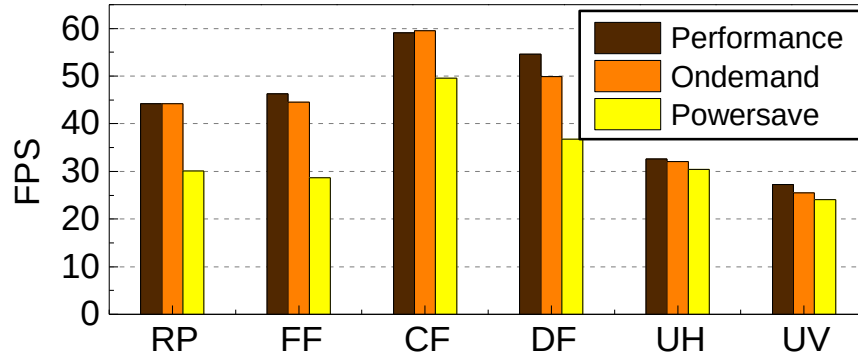
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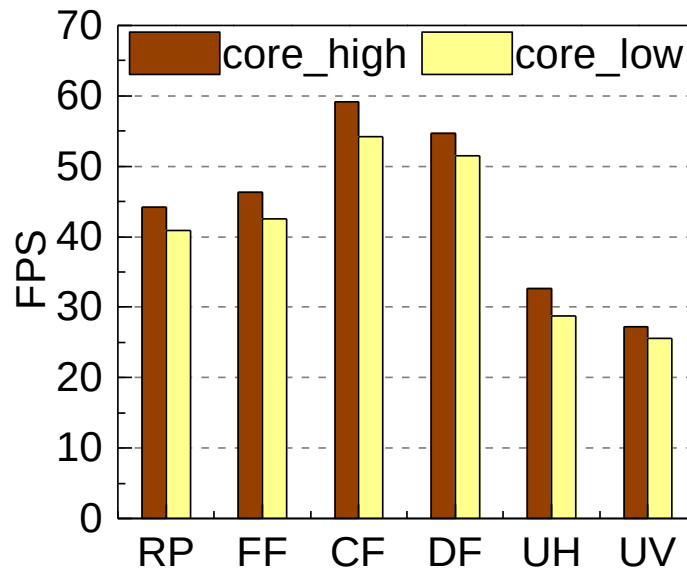
DVFS-CPU



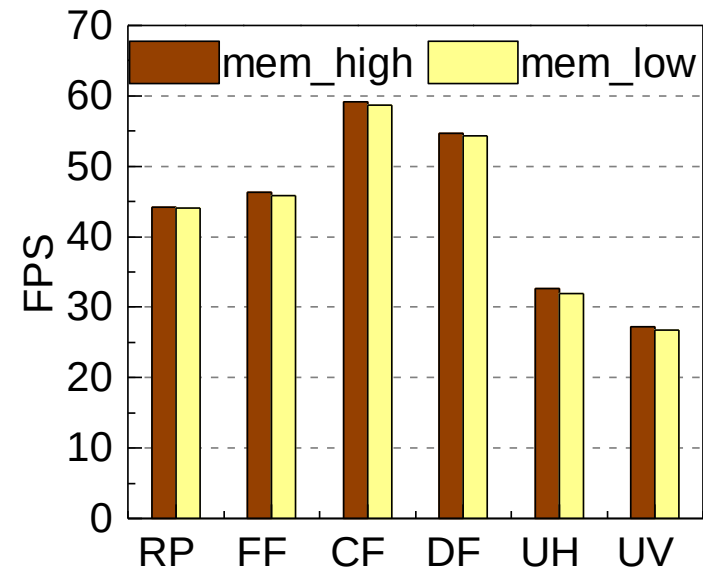
- Remarks:

Ondemand CPU frequency scaling achieves the best performance tradeoff between performance and energy for GaaS

Core 745→575Mhz



Mem 1250→750Mhz



- Remarks:

The GPU memory frequency can be tuned lower within a certain range to get energy saving with little performance degradation for GaaS.

Conclusions



- In this work, we conduct a characterization on XenServer using virtual GPU, we found no NUMA overhead for GaaS workloads, due to the fact that most memory copy operations are overlapped with GPU computation.
- GaaS workloads exhibits different workload behavior with GPGPU workloads.
- Ondemand CPU frequency scaling achieves the best tradeoff between performance and energy for GaaS.
- GPU memory clock can be tuned lower within a certain range to save energy for GaaS.

Thanks For Your Attention!

